Mohammad Sami Khan

AI/ML ENGINEER

AI/ML Engineer with end-to-end projects spanning real-time computer vision (YOLO, SegFormer) and generative AI (RAG, LangChain). Experienced in fine-tuning models, evaluating performance, and building deployment-ready pipelines with PyTorch and FastAPI.

TECHNICAL SKILLS

COMPUTER VISION & REAL-TIME SYSTEMS

YOLO, OpenCV, Object Detection & Tracking, Semantic Segmentation, Video Analytics, Image Classification

GENERATIVE AI & LLM APPLICATIONS

RAG, LangChain, Embedding Models, ChromaDB, Google Gemini API, Prompt Engineering

DEEP LEARNING & MODEL DEVELOPMENT

PyTorch, CNNs, Transfer Learning, Model Fine-Tuning, Evaluation Metrics

DEPLOYMENT & ENGINEERING

FastAPI, Hugging Face Transformers, Streamlit, NumPy, Pandas, Matplotlib, sklearn

PROGRAMMING & TOOLS

Python, Git/GitHub, SQL

CONTACT

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github.com/Samikhxnn

LOCATION

delhi,india

PROJECTS

SMART CCTV ANALYTICS SYSTEM

Python • PyTorch • YOLOv11 • OpenCV • Real-Time Object Detection & Tracking

- Developed a real-time video analytics system using Python, PyTorch, YOLO, and OpenCV for person detection, multi-object tracking, crowd monitoring, fire/smoke detection, and intrusion detection.
- Integrated automated Telegram alerts with notifications for security events.
- Designed a modular pipeline supporting multiple computer vision tasks from a single video stream and optimized the inference pipeline for efficient real-time processing.

Github | Demo

COST-EFFICIENT RAG QA SYSTEM

Python • LangChain • ChromaDB • Google Gemini • FastAPI

- Built a RAG-based document question answering system using LangChain and ChromaDB.
- Implemented PDF ingestion, chunking, embeddings, and semantic vector search.
- Developed a FastAPI service with grounded responses and source citations.
- Evaluated retrieval quality, latency, and cost using standard RAG metrics.

Github

AUTONOMOUS DRIVING SEMANTIC SEGMENTATION USING SEGFORMER

Python • PyTorch • Hugging Face Transformers • SegFormer • OpenCV

- Fine-tuned a pretrained SegFormer model using PyTorch for multi-class semantic segmentation on urban driving scenes.
- Implemented custom dataset loading, preprocessing, augmentation, evaluation with Mean IoU, and inference for pixel-level scene segmentation.

Github | Demo

REAL TIME HAND SIGNS DETECTION

Python • PyTorch • YOLOv11 • OpenCV

- Developed a real-time hand sign detection system using YOLOv11, PyTorch, and OpenCV for live gesture recognition from webcam input.
- Implemented real-time video processing, object detection, and visualization pipeline for accurate hand sign classification.

Github | Demo

EDUCATION

DR.ISLAM MAJEED COLLEGE , BSC 2024-2027

BSC CGPA In progress

Bargain,Etah, Uttar Pradesh

BLOOMING DALE SCHOOL INTERMEDIATE 2020-2021

CBSE Percentage 88 %

Budaun, Uttar Pradesh

DE PAUL SCHOOL MATRICULATE 2018-2019

ICSE Percentage 89.5 %

Budaun, Uttar Pradesh

COURSE $ CERTIFICATION

TECH STACK ACADEMY,DELHI AIML COURSE 2025- 2026

- Artificial Intelligence & Machine Learning Certification
- project based learning

FREECODECAMP,ONLINE

DEEP LEARNING $ PYTORCH 2025

LANGUAGES

- English - Working Proficiency
- Hindi - Native Proficiency
- Urdu - Native Proficiency